

A token selection-based multi-scale dual-branch CNN-transformer network for 12-lead ECG signal classification

Siyuan Zhang^a, Cheng Lian^{a,*}, Bingrong Xu^a, Junbin Zang^b, Zhigang Zeng^c

^a School of Automation, Wuhan University of Technology, Wuhan 430074, China

^b School of Instrument and Electronics, North University of China, Taiyuan 038507, China

^c School of Artificial Intelligence and Automation, Huazhong University of Science and Technology, Wuhan 430074, China

ARTICLE INFO

Article history:

Received 28 June 2023

Received in revised form 1 September 2023

Accepted 13 September 2023

Available online 16 September 2023

Keywords:

12-lead ECG signal classification

Convolutional neural network

Transformer

Multi-scale learning

Token selection

Information redundancy

ABSTRACT

The timely identification of cardiovascular diseases is critical for effective intervention, with the electrocardiogram (ECG) serving as a pivotal diagnostic tool. Recent advancements in deep learning-based methods have significantly enhanced the accuracy of ECG signal classification. In clinical settings, cardiologists rely on diagnoses derived from standardized 12-lead ECG recordings. It must be acknowledged that there is considerable redundancy in the 12-lead ECG recordings used for ECG signal classification, thereby hindering their generalization capabilities. Meanwhile, considering multi-scale features in 12-lead ECG recordings is a crucial aspect that is often overlooked by existing methods. Based on the above observations, we develop a multi-scale Convolutional Transformer network for ECG signal classification. By utilizing learnable Convolutional neural network (CNN) blocks and novel dual-branch Transformer encoders, the proposed network automatically extracts features at different scales, resulting in superior feature representation. Additionally, by discarding low-importance patches and focusing on high-importance patches, we effectively alleviate information redundancy in the 12-lead ECG recordings. We conduct comprehensive experiments on three commonly used ECG datasets. The Research results show that our proposed network outperforms existing state-of-the-art networks in multiple tasks.

© 2023 Elsevier B.V. All rights reserved.

1. Introduction

Cardiovascular disease stands as one of the most formidable threats to human health globally, claiming a staggering number of lives each year. In 2017, this pervasive condition accounts for an estimated 17.8 million deaths worldwide, reflecting its status as a leading cause of mortality [1]. Therefore the significance of detecting and treating cardiovascular disease has garnered increasing attention worldwide.

A 12-lead ECG signal recording has emerged as a pivotal diagnostic tool for the screening and evaluation of cardiac abnormalities. This cost-effective and noninvasive detection method holds immense value, as it offers a comprehensive assessment of the electrical activity of the heart. By providing a wealth of information, the 12-lead ECG aids in the diagnosis of various abnormalities, enabling timely intervention and treatment. However, detecting abnormal ECG signal poses significant challenges and demands considerable time and effort from cardiologists. Therefore, the rapid advancement of smart medicine has opened

up new avenues for assisting cardiologists in the automatic ECG signal classification [2].

Machine learning methods, such as support vector machines, random forests, etc., have been extensively employed for ECG signal classification [3]. While these methods have shown promise, their accuracy still falls short compared to that of cardiologists. This discrepancy can be attributed to the inherent complexity of classification tasks and the limited feature extraction capabilities of traditional machine learning methods. Traditional machine learning methods for ECG signal classification typically involve three key steps: preprocessing, feature extraction, and classification. Among these steps, feature extraction plays a crucial role in determining the accuracy of the overall system. However, the reliance on manual feature construction and the need for specialized expertise have posed significant challenges, impeding the desired accuracy in ECG signal classification.

In recent years, deep learning-based methods for ECG signal classification have made remarkable strides, achieving cardiologist-level performance. Unlike traditional machine learning methods, these methods can analyze ECG signal without the need for manual feature extraction, revolutionizing the field of ECG signal classification. Hannun et al. [4] propose a 34-layer CNN capable of accurately classifying a wide range of arrhythmias from

* Corresponding author.

E-mail address: chenglian@whut.edu.cn (C. Lian).

single-lead ECG signal, achieving high diagnostic performance comparable to that of cardiologists. Schwab et al. [5] develop an ensemble of recurrent neural networks (RNNs) with an attention mechanism to classify heart rhythms in ECG signal, which achieves state-of-the-art performance. Pengwei et al. [6] propose a novel ECG beat classification algorithm, which combines a Bidirectional RNN for capturing temporal dependencies with a CNN for morphological feature extraction. The Transformer, a widely adopted deep neural network architecture, is also leveraged for ECG signal classification [7].

Although deep learning-based methods have demonstrated significant advancements in improving the accuracy of ECG signal classification, there are still many directions to explore and unsolved challenges in this field. In the field of computer vision, multi-scale feature learning play a crucial role in various vision tasks [8]. When it comes to ECG signals, different time scales exhibit distinct characteristics. Particularly, a 12-lead ECG signal recording often contains multiple leads and each lead is composed of many sub-beat sequences. Therefore, to fully understand ECG signal, extracting features at different time scales is necessary for ECG signal classification networks. Meanwhile, the 12-lead ECG signal recording is widely utilized in clinical practice as it offers a comprehensive assessment of cardiac electrical activity. However, when training deep learning-based networks using the complete 12-lead ECG signal recording as input, a common issue is the presence of information redundancy [9]. Therefore, deep learning-based methods stand to gain significant improvements in generalizability by addressing inherent redundancy in the 12-lead ECG signal recording.

In consideration of the above discussion, this paper investigates how to utilize multi-scale feature learning to obtain more multi-scale feature information of 12-lead ECG signal recordings and effectively solve inherent redundancy in the 12-lead ECG signal recording. **The main contributions of this paper are summarized as follows.**

- We propose a multi-scale dual-branch CNN-Transformer network, designed to extract and merge multi-scale global and local features, which can achieve a more comprehensive understanding of the ECG signal and improve classification performance.
- We adopt a dual-branch structure using a multi-scale embedding layer (MSEL) which results in the generation of cross-scale tokens. The MSELs in both branches can extract features at different scales of the ECG signal.
- We propose a token selection block (TS block) to preserve high attention tokens and discard low attention tokens that contribute little to the final prediction, which can reduce the negative impact of redundant information.
- We introduce a feature correlation loss term in our loss function which suppresses redundant information and enhances the discriminative power of extracted features.
- Experiments on three ECG datasets demonstrate that our proposed network outperforms state-of-the-art networks on various classification tasks.

The rest of the paper is organized as follows. In Section 2, we introduce the related studies on machine learning methods, and deep learning-based methods for ECG signal classification. In Section 3, we present the details of the proposed network. Section 4 reports the research design. Section 5 reports the comparative analysis with previous methods, and performs ablation experiments. Finally, in Section 6, we conclude this paper and provide directions for future work.

2. Related work

The classification of ECG signal using deep learning techniques is a highly challenging and impactful research area. Over the years, many methods have been proposed to tackle this task. In this section, we explore the application of traditional machine learning methods and deep learning-based methods to the task of ECG signal classification.

2.1. Machine learning methods for ECG signal classification

Over the past few decades, a great deal of research has been devoted to the classification and diagnosis of heart disease using traditional machine learning methods. Jha et al. [10] propose a tunable Q-wavelet transform to efficiently extract features from ECG signal and then use the support vector machine as a classifier to perform the pattern recognition of cardiac arrhythmias. Jose et al. [11] first use wavelet transform to extract features of ECG signal and feed features to the probabilistic neural network to perform classification. Mariano et al. [12] use a patient-adaptable algorithm for ECG signal classification, which combines a linear discriminant classifier and an expectation maximization clustering algorithm to perform efficient feature classification.

However, the traditional machine learning methods for ECG signal classification faces several problems. Pre-processing of raw ECG signal and feature engineering in the traditional machine learning methods for ECG signal classification can be time-consuming and labor-intensive. The selection and extraction of relevant features from the ECG signal require domain expertise and manual intervention, which can introduce subjective biases and limit the scalability of the classification approach. Furthermore, the traditional machine learning methods often rely on artificially crafted features, which may not capture the full complexity and variations present in ECG signal. These artificially crafted features may not be optimized for specific classification tasks or may fail to capture subtle patterns and characteristics that are important for accurate classification. Moreover, the traditional machine learning methods may struggle to handle noise and artifacts, leading to poor classification performance and increasing the possibility of misclassification errors.

2.2. Deep learning-based methods for ECG signal classification

Recently, with the extensive application and success of deep learning-based methods in various fields, researchers have proposed many deep neural network architectures for ECG signal classification. Time sequence networks such as long short-term memory (LSTM) [13], gated recurrent unit (GRU) [14], and variants based on RNNs have gained significant popularity due to their ability to capture temporal dependencies in ECG signal. CNNs have also been widely used in ECG signal classification. Murugesan et al. [15] propose a robust deep neural network architecture, named ECGNet, for automated feature extraction and classification of ECG signals, outperforming existing methods in ECG signal classification. Pu et al. [16] propose an ultralightweight binary neural network (BNN) for ECG classification, achieving high accuracy with minimal storage usage. Yao et al. [17] propose an attention-based time-incremental convolutional neural network (ATI-CNN) for automatic arrhythmia detection from ECG signals, achieving flexible input length, improved performance in detecting paroxysmal arrhythmias, and significant computation reduction. Chen et al. [18] propose a convolutional neural network model, named ACNet, that achieves best performance in the China Physiological Signal Challenge (CPSC) 2018 by accurately detecting and classifying cardiac arrhythmias. Yang et al. [19] propose a multi-view and multi-scale deep neural

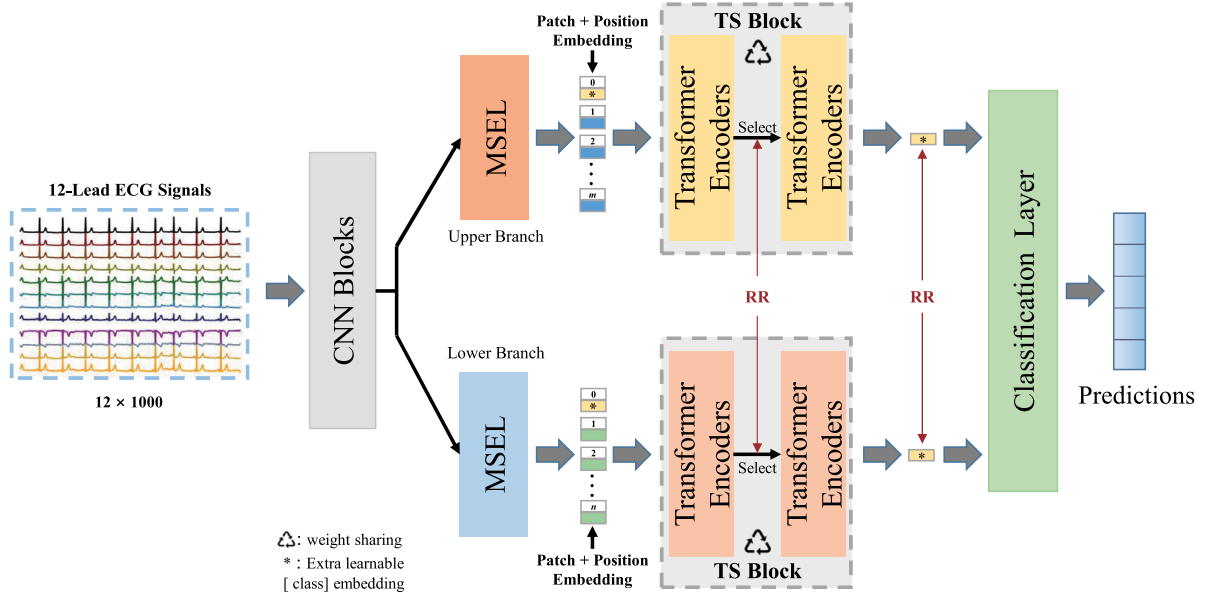


Fig. 1. Overview of our network. Our network processes 12-lead ECG signals using a dual-branch structure. The signal is first processed using CNN blocks, and then each branch obtains patches using MSEL components of different scales. These patches are input to the Transformer encoders for feature extraction, and patches with high information are selected to be input to the next encoders for extraction. Finally, a classification layer is used to make classification predictions. RR represents calculating the correlation coefficient between the CLS tokens in the two branches. Weight sharing between two sets of Transformer encoders in the same TS block. m and n represent the number of tokens in each branch respectively.

network for ECG signal classification, utilizing the diversity of different lead features in a 12-lead signal ECG. This network outperforms existing state-of-the-art networks in multiple tasks. Motivated by the multi-instance learning on histopathological image, Han et al. [20] propose a novel ECG heartbeat classification method based on multi-instance learning, which effectively captures rhythmic patterns in ECG signal.

Our proposed network first proposes to use Transformer-based multi-scale dual-branch structure to extract multi-scale feature information in ECG signal classification. Compared to state-of-the-art methods, our proposed network obtains almost the best results on three commonly used ECG datasets.

3. Methods

In this section, we first introduce the overall structure of our proposed network. Then, the components of our network are introduced in detail. Specifically, the proposed network primarily includes four parts: a CNN block, a multi-scale embedding layer (MSEL) for dual-branch, a token selection (TS) block, and the feature correlation loss term $\mathcal{L}_{FC}$.

3.1. Method overview

The overall architecture of our proposed network is outlined in Fig. 1. Our proposed network employs a dual-branch structure, which naturally splits 12-lead ECG signal into patches of different scales. We first process the 12-lead ECG signal using CNN blocks. Then each branch gets patches of different sizes using MSEL components with different scales to process the data from CNN blocks. The upper branch samples patches using two kernels with a large difference in size, and the lower branch samples patches using two kernels with a small difference in size. All the tokens from both branches are fed into their respective Transformer encoders to extract features. These encoders

Table 1

The parameter setting of the CNN block.

	Filter	Stride	Padding	Out_channels
Conv1	7	1	3	128
Conv2	5	1	2	256
Conv3	3	1	1	128
Conv4	1	1	0	32
Conv5	1	1	0	32
Conv6	39	1	19	32
Conv7	19	1	9	32
Conv8	9	1	4	32
Conv9	1	1	0	128

utilize self-attention mechanisms to capture important patterns and relationships within the tokens. After the initial feature extraction, the high informative tokens from both branches are selected based on their significance and forwarded to the next set of Transformer encoders. Weight sharing between two sets of Transformer encoders in the same TS block. Meanwhile, by calculating the correlation coefficient between the CLS tokens in the two branches, we introduce the feature correlation loss term $\mathcal{L}_{FC}$ in our loss function. To perform the classification task, a classification layer is introduced as the final layer of the network, responsible for predicting the probability scores for each class.

3.2. CNN block

The structure of the CNN block is shown in Fig. 2, and the parameter settings of each convolutional layers are outlined in Table 1. The CNN block is modified on the basis of Inception-Time block [21]. The CNN block uses a multi-scale structure that effectively captures ECG signal features at different time scales. Notably, ECG signal features obtained at different scales exhibit distinctive patterns of change, reflecting unique trends within the signal. By leveraging the multi-scale structure, the CNN block

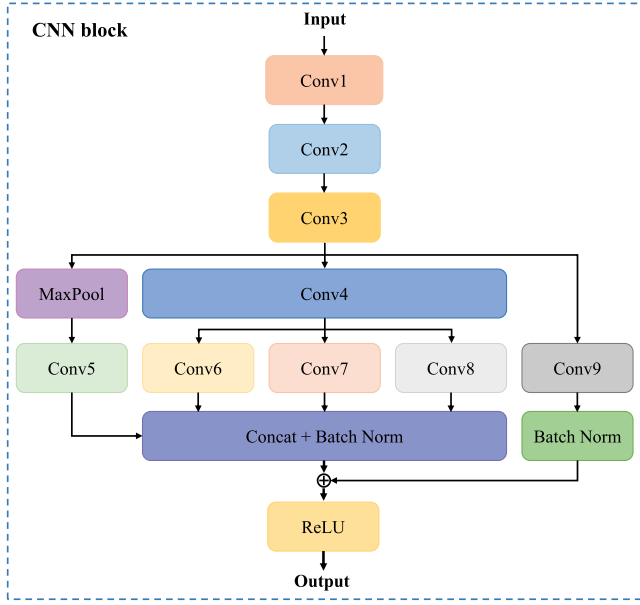


Fig. 2. The structure of the CNN block. The CNN block in our network employs a multi-scale structure to capture ECG signal features at different time scales. $\oplus$ is element-wise addition.

can capture and utilize these diverse features, enhancing the performance of the overall network in ECG signal classification.

3.3. Multi-Scale Embedding Layer (MSEL)

Patch Embedding is a commonly used method to reshape image input into a sequence of flattened 2D patches, which can be the input of the standard Transformer. To address the challenge of processing large images with Transformer, a compromise is made by dividing the image into multiple smaller patches. These patches are linearly embedded, which means that they are transformed into lower-dimensional representations. Transformer is then applied directly to this collection of patches instead of processing the entire image as a whole. It is proposed in [22] that the patch-based representation using the patch embedding plays a crucial role in the excellent performance of Transformer.

In ECG signal classification, the distinctive waveform characteristics associated with different cardiac conditions or symptoms are essential for accurate diagnosis. These waveform patterns, such as P-waves, QRS complexes, and T-waves, provide valuable information for identifying specific abnormalities or symptoms. However, many existing methods preprocess ECG signals by segmenting them into individual beat segments alone, overlooking the correlation between beat segments within a specific time scale. To address this limitation, we propose the use of a multi-scale embedding layer (MSEL) to extract cross-scale features from ECG signals. Meanwhile, we use a multi-scale dual-branch structure. The multi-scale features extracted from the upper branch and the lower branch have distinct properties, which enables the network to learn mutually complementary information from different time scale.

The structure of MSEL is shown in Fig. 3. MSEL is leveraged to obtain input tokens for each branch Transformer encoders. MSEL introduces a novel strategy to ensure that an equal number of tokens are obtained from two 1D kernels of different sizes. By maintaining the same stride for both kernels, the resulting patches share the same center point on the input data. These two patches are then embedded into the same dimension and concatenated into a single embedding token, which preserves

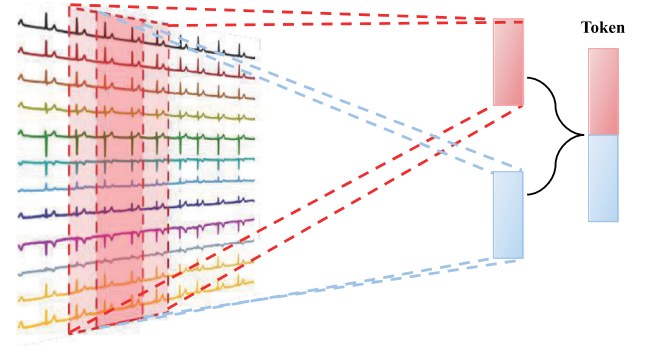


Fig. 3. Illustration of the MSEL layer. To sample the input data, two distinct kernels with an identical stride are employed. The embedding process involves constructing embeddings for each patch and subsequently concatenating the two patches together.

the spatial relationship between the different scales. Meanwhile, in the lower branch, we use a large 1D kernel and a small 1D kernel (1×5 and 1×100) in the MSEL to fuse broader feature information and local feature information. In the upper branch, we use two middle 1D kernels (1×20 and 1×40) in the MSEL to capture feature information at intermediate scales. By considering the correlation information among beat segments at different time scales, the MSEL of dual-branch can better capture the cross-scale features.

3.4. Token selection block

Transformer encoder is a powerful structure for capturing long-range dependencies in patch sequences. In the following, we describe the Transformer encoder for the upper branch, and the same procedure is performed for the lower branch. With the help of MSEL, the output data from CNN blocks is divided into a token sequence. An additional classification (CLS) token and learnable position embeddings are added to the token sequences for the two branches, as in ViT [23]. Thus, the input token sequence of the Transformer encoder is:

$$\mathbf{X} = [\mathbf{x}^c; \mathbf{x}^1; \dots; \mathbf{x}^m] + \mathbf{E}_{pos} \quad (1)$$

where $\mathbf{x}^c \in \mathbb{R}^D$ is the CLS token and $\mathbf{x}^i$ is the i th patch tokens. The $\mathbf{E}_{pos} \in \mathbb{R}^{m \times D}$ is the set of position embeddings, m is the number of patch tokens, and D is the dimension of the embedding tokens.

The structure of the Transformer encoder is shown in Fig. 4, which consists of a multi-headed self-attention (MSA) and a multilayer perceptron (MLP) block. Given the input token sequence $\mathbf{X}$, the self-attention operation in the MSA calculation is as follows:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}(\mathbf{Q}\mathbf{K}^T / \sqrt{D})\mathbf{V} \quad (2)$$

The query matrix $\mathbf{Q} \in \mathbb{R}^{(m+1) \times D}$, key matrix $\mathbf{K} \in \mathbb{R}^{(m+1) \times D}$, and value matrix $\mathbf{V} \in \mathbb{R}^{(m+1) \times D}$ are calculated by $\mathbf{X}$ through transformation matrices. Then the output tokens from the MSA are sent to the MLP block. Finally, the CLS token is taken out for ECG signal classification in the last Transformer encoder.

Long-term ECG signal contains many heartbeats, but the proportion of arrhythmic heartbeats within these signals is typically very low. Therefore, detecting and identifying arrhythmic heartbeats among these normal heartbeats is a challenging task in ECG signal classification. Meanwhile, many instances of redundancy are observed within the ECG signals of normal heartbeats, rendering them inconsequential for the classification outcome. Based on these arguments, we propose the TS block which can identify and remove these tokens that are no helpful for classification performance. The process is shown in Fig. 5.

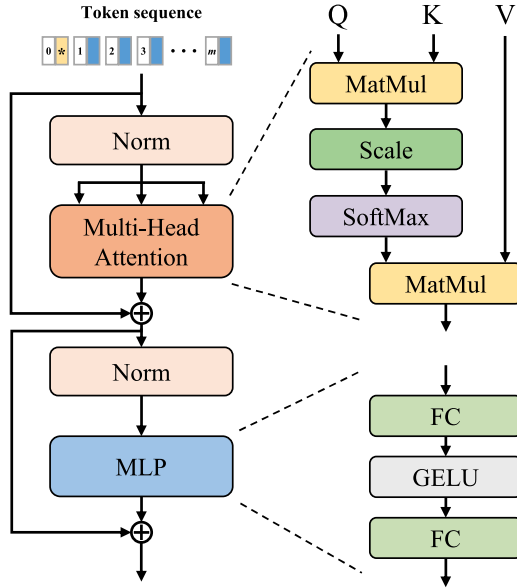


Fig. 4. The structure of the Transformer encoder, where FC is a fully connected layer and $\oplus$ is element-wise addition. $[\ast]$ is the classification token.

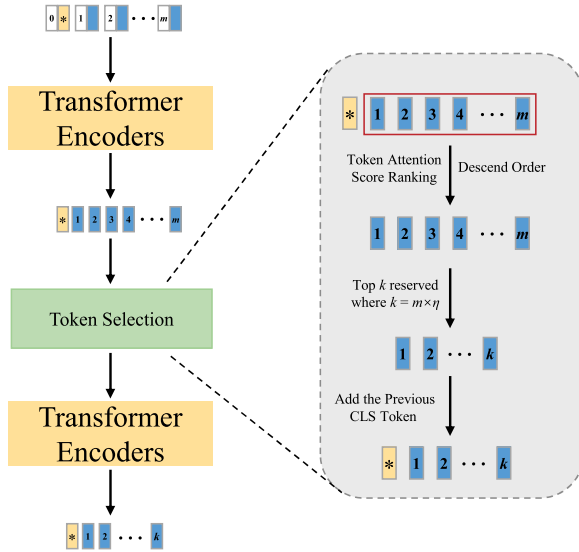


Fig. 5. The structure of the TS block. The Transformer encoders reprocess the patches, selecting those with significant information for another round of feature extraction. Weight sharing is employed between two sets of Transformer encoders within the same TS block.

For the CLS token extracted from the last encoder, the interactions with other patch tokens are expressed as:

$$\mathbf{x}^c = \text{Softmax}(\mathbf{q}_{\text{cls}} \mathbf{K}^T / \sqrt{D}) \mathbf{V} = \mathbf{a} \mathbf{V} \quad (3)$$

where $\mathbf{q}_{\text{cls}}$ denotes the query vector of the CLS token, $\mathbf{a} = [a_0; a_1; \dots; a_m]$ is the combination coefficient, and the attention value a_j represents the importance of the j th token to the CLS token $\mathbf{x}^c$. Therefore, it represents that if a_j is lower, the corresponding token is less informative about the CLS token.

To stabilize the class attention and reduce the impact of inconsistent attention values in the bottom layers, we use the global class attention [24] based on exponential moving average (EMA):

$$\hat{\mathbf{a}}_l = \lambda \mathbf{a}_{l-1} + (1 - \lambda) \mathbf{a}_l, \quad l = 1 \dots L \quad (4)$$

where L represents the total number of the Transformer encoder and $\lambda = 0.99$. The global class attention of the upper branch and the lower branch starts from the 4th encoder and the 3th encoder, respectively. Depending on the global class attention, we select a subset of patch tokens based on their attention scores. Specifically, we choose the top- k patch tokens with the highest attention scores among the m patch tokens, where $k = m \times \eta$ and $0 \leq \eta \leq 1$. Then, we put the selected high-importance patch tokens into the next set of Transformer encoders for feature extraction again. Weight sharing is employed between two sets of Transformer encoders within the same TS block. By iteratively extracting features from the high informative patches, our proposed network can capture and emphasize the most relevant information.

3.5. Classification layer

Individual CLS tokens from two branches are extracted and passed to the classification layer to undertake the classification task. These two CLS tokens are combined to build a vector, which is then subjected to a series of operations in the classification layer. The classification layer consists of a linear layer and an activation layer. The final result of the classification layer is to generate a probability score for each class as the model's predicted output.

3.6. Training loss function

R-Squared represents the degree to which independent variable $\mathbf{x}$ explains the dependent variable $\mathbf{y}$ in a linear regression network, and can also be understood as an indicator of the fit of the linear regression network. The range of R-Squared values is $[0, 1]$. If R-Squared = 1, it indicates that the network is fully fitted, and the independent variable $\mathbf{x}$ can completely explain the change in the dependent variable $\mathbf{y}$. The closer the R-square is to 1, the higher the fit of the network. The formula for R-Squared is as follows:

$$R^2(\mathbf{x}, \mathbf{y}) = \frac{\sum_{i=1}^B (x_i - \bar{y})^2}{\sum_{i=1}^B (y_i - \bar{y})^2} = 1 - \frac{\sum_{i=1}^B (y_i - x_i)^2}{\sum_{i=1}^B (y_i - \bar{y})^2}, \quad (5)$$

where $\mathbf{x} = [x_1; x_2; \dots; x_B]$, $\mathbf{y} = [y_1; y_2; \dots; y_B]$ and $\bar{y} = \frac{\sum_{i=1}^B y_i}{B}$. $\sum_{i=1}^B (y_i - x_i)^2$ represents the sum of the squared differences between the observed values of $\mathbf{y}$ and the predicted values obtained from the linear regression network, and $\sum_{i=1}^B (y_i - \bar{y})^2$ represents the sum of the squared differences between the observed values of $\mathbf{y}$ and their mean value.

In the training stage, the loss function of our proposed network can be defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}(\mathbf{Y}, \text{CLASS}) + \alpha \mathcal{L}_{\text{FC}}, \quad (6)$$

where $\mathcal{L}(\mathbf{Y}, \text{CLASS})$ is the loss of computing the loss value between the predicted value $\mathbf{Y}$ and the ground truth label CLASS, $\mathcal{L}_{\text{FC}}$ is the feature correlation loss term, and α is the tuning parameter. For the CPSC dataset [25] and the PTB-XL dataset [26] as the multi-label datasets, we use BCE with logits loss as $\mathcal{L}(\mathbf{Y}, \text{CLASS})$. For the Chapman dataset [27] as the single-label datasets, we use the cross-entropy loss function as $\mathcal{L}(\mathbf{Y}, \text{CLASS})$.

Additionally, our proposed feature correlation loss term $\mathcal{L}_{\text{FC}}$ is described as:

$$\mathcal{L}_{\text{FC}} = \frac{\mathcal{L}_{\text{RR}}^A}{\mathcal{L}_{\text{RR}}^B} = \frac{R^2(\hat{\mathbf{x}}_{\text{upper}}^c, \hat{\mathbf{x}}_{\text{lower}}^c)}{R^2(\mathbf{x}_{\text{upper}}^c, \mathbf{x}_{\text{lower}}^c)}, \quad (7)$$

where $R^2(\cdot)$ is the R-Squared operator, $\mathbf{x}_{\text{upper}}^c$ and $\mathbf{x}_{\text{lower}}^c$ are respectively the CLS tokens extracted from the output token sequences of the first set of Transformer encoders on the dual-branch TS block, and $\hat{\mathbf{x}}_{\text{upper}}^c$ and $\hat{\mathbf{x}}_{\text{lower}}^c$ are the CLS tokens extracted from the

Table 2

Number of ECG recordings for training set, validation set and test set in each dataset.

Dataset	# Classes	# Train	# Val	# Test	# Total
CPSC	9	5497	690	690	6877
PTB-XL	19	7202	904	882	8988
Chapman	4	8516	1065	1065	10646

selected output token sequences of the next set of Transformer encoders of the two branches, respectively. The feature correlation loss term is designed to encourage correlation between features extracted from the two branches. By measuring the similarity or dissimilarity between feature representations, we can guide the network to focus on relevant and discriminative information while discouraging the presence of redundant or irrelevant features.

This feature correlation loss term is motivated by the fact that $\hat{\mathbf{x}}_{\text{upper}}^c$ and $\hat{\mathbf{x}}_{\text{lower}}^c$ contain different time scale features and the same redundant information. Under the guidance of $\mathcal{L}_{\text{FC}}$ in gradient descent, $\mathcal{L}_{\text{RR}}^B$ gradually becomes larger, which can pull away individual different time scale features. In contrast, $\hat{\mathbf{x}}_{\text{upper}}^c$ and $\hat{\mathbf{x}}_{\text{lower}}^c$ are input into the classifier for the classification task. Thus, they are often highly correlated. Under the guidance of $\mathcal{L}_{\text{FC}}$ in gradient descent, $\mathcal{L}_{\text{RR}}^A$ gradually becomes smaller, which can obtain consistent representation with these two CLS tokens. By introducing this loss term, our total loss function can effectively alleviate redundant information in consistent representation, and improve the quality of the extracted features.

4. Research design

4.1. Datasets

We employ three datasets to comprehensively evaluate our proposed network. These three datasets have been widely used to evaluate the performance of various ECG-based classification algorithms. We use the same scale split for all three datasets. For each dataset, we split into 10 groups. Groups 1 to 8 are used as the training set to train the network and learn latent patterns in the data. Group 9 is used as a validation set for network selection and hyperparameter tuning. Finally, group 10 is designated as the test set as an independent evaluation set to evaluate the performance of the network on unseen data. Table 2 shows the number of ECG recordings in different partition sets on these three datasets, and also includes the number of categories for each and the number of overall ECG recordings.

4.1.1. CPSC dataset

The China Physiological Signal Challenge 2018 (CPSC 2018) dataset [25] is a multi-label ECG dataset where each 12-lead ECG recording ranges from 6 to 60 s and is sampled at a frequency of 500 Hz. To reduce the computational burden, the sampling frequency is reduced from 500 Hz to 100 Hz. To ensure consistent data lengths for analysis, the longer ECG recordings are cropped to 10 s, while the shorter ones are upsampled to reach the same 10-s duration. Then the dataset is normalized. By applying these steps, the dataset is standardized to contain ECG recordings of exactly 10 s in length.

4.1.2. PTB-XL dataset

The PTB-XL dataset [26] is a multi-label ECG dataset that contains a large number of 12-lead ECG recordings. It obtains these 12-lead ECG recordings from 18,885 patients. This ECG dataset encompasses a total of 71 distinct ECG statements, which can be categorized into 44 diagnostic statements (diag.), 19 formal statements (form), and 12 rhythmic statements (rhythm).

Considering both economy and efficiency as important factors, we have selected the form task for our research. The form task encompasses 8988 ECG recordings of 10 s length. The original ECG signals in the PTB-XL dataset are sampled at two different frequencies: 500 Hz and 100 Hz. For our study, we have chosen a sampling frequency of 100 Hz. This frequency allows us to capture essential information from the ECG signal while reducing the computational burden. To ensure consistency and comparability across different recordings, we applied data normalization to the dataset.

4.1.3. Chapman dataset

The Chapman dataset [27] is a single-label ECG dataset that provides 12-lead ECG recordings from 10,646 patients. Each ECG recording in the dataset has a fixed duration of 10 s. The original sampling frequency of the Chapman dataset is 500 Hz. However, for the purpose of this study, the dataset has been downsampled to a frequency of 100 Hz. In addition to downsampling, data normalization is applied to the Chapman dataset. This step ensures that the data is on a consistent scale and facilitates fair comparisons and accurate analysis.

4.2. Introduction of the comparison network

(1) Baseline Deep Networks for Time Series Classification:

(1) fcn_wang [28]: This network consists of three convolution layers and a global average pooling layer. The structure of fcn_wang is simple but effective. (2) resnet1d_wang [28]: This network is a kind of ResNet [29] which uses three residual blocks and a global average pooling layer. (3) resnet1d101 [30]: The network utilizes an adapted ResNet which stacks multiple layers of residual blocks and bottleneck blocks. (4) InceptionTime [31]: InceptionTime is a customized architecture based on InceptionV4 [32] for time series analysis and consists of six improved Inception modules. These modules consist of parallel branches with different sizes to capture features at different time scales. (5) MobileNetV3 [33]: MobileNetV3 is an efficient and lightweight neural network which combines depthwise separable convolutions and linear bottlenecks for high accuracy. (6) LSTM [34]: Long Short-Term Memory is a type of RNN that is designed to address the limitations of traditional RNNs when dealing with long sequences. LSTM can effectively retain and update long-term information, avoiding the vanishing or exploding gradient problem. (7) BiLSTM [35]: Bidirectional Long Short-Term Memory is an extension of the LSTM architecture that introduces bidirectional processing to capture the semantic dependencies in both directions. (8) ViT [23]: The Vision Transformer (ViT) is a Transformer architecture for computer vision tasks. It consists of multiple layers, each containing a self-attention mechanism and a feed-forward network.

(2) State-of-The-Art Networks for ECG Signal Classification:

(1) ECGNet [15]: This network is a ECG signal classification network which combines CNN and LSTM. It firstly uses the LSTM architecture for ECG classification and gets the improved accuracy. (2) ECG_BNN [16]: It is a lightweight network which is easy to program for hardware. It removes the fully connected layers and is the state-of-the-art in known lightweight networks. (3) ATI-CNN [17]: This network combines a fully convolutional neural network inspired by VGGNet, LSTM layers, and an attention module to enable spatial fusion of information from different ECG leads and enhance feature representation. (4) AC-Net [18]: This network consists of five CNN blocks, followed by a bidirectional RNN layer, an attention layer, and a Dense layer. (5) MVMSNet [19]: This network is a deep neural network based on multi-view learning. It uses the coordinate attention and employs a multi-scale CNN to get different useful features for ECG classification.

Table 3

Architectural configurations and experiment configurations. U-B and L-B are the abbreviations of upper branch and lower branch respectively. LR means the initial learning rate.

Network hyper-parameter															Training process			
CNN blocks		U-B MSEL	L-B MSEL	U-B TS block					L-B TS block					LR	Batch size	Epochs	α	
Layers	Kernel size & Stride	Kernel size & Stride	Layers	Heads	Dim	Ratio	η	λ	Layers	Heads	Dim	Ratio	η					λ
2	(20, 40), 20	(5, 100), 5	8	8	424	4	0.5	0.99	5	8	424	4	0.5	0.99	1×10^{-3}	64	200	0.2

Table 4

Our proposed network is compared with baseline deep networks for time series classification and ECG signal classification networks for the classification tasks of the CPSC dataset, the PTB-XL dataset, and the Chapman dataset. Bold numbers indicate the best performance. Underscore indicates the second best performance.

Method	CPSC				PTB-XL				Chapman			
	F ₁	AUC	Recall	mAP	F ₁	AUC	Recall	mAP	F ₁	AUC	Recall	Accuracy
fcn_wang [28]	69.80	91.85	75.58	84.36	23.46	79.07	51.32	69.56	83.88	93.95	83.26	85.54
resnet1d_wang [28]	75.52	94.53	80.36	87.28	29.75	83.73	54.70	71.01	92.47	98.40	92.63	93.05
resnet1d101 [30]	79.78	95.64	<u>85.94</u>	<u>90.60</u>	31.12	82.59	52.92	70.16	95.00	99.14	94.98	<u>95.40</u>
LSTM [34]	72.41	93.71	78.84	87.56	22.13	83.17	53.87	70.84	91.79	97.53	91.82	92.58
BiLSTM [35]	76.65	<u>96.04</u>	81.52	89.19	23.50	83.94	51.32	69.42	93.64	98.53	93.75	94.18
InceptionTime [31]	76.84	93.61	82.25	87.33	<u>32.32</u>	<u>85.54</u>	56.23	72.37	92.20	98.14	92.50	92.96
ViT [23]	64.89	89.82	69.35	78.46	18.98	75.55	42.43	61.20	81.60	95.72	82.06	83.19
MobileNetV3 [33]	75.64	94.33	79.35	85.91	20.07	74.96	20.07	49.53	94.00	99.08	94.08	94.55
ECGNet [15]	79.49	95.11	82.68	88.31	27.24	83.30	54.89	72.29	<u>95.98</u>	<u>99.39</u>	<u>96.07</u>	95.02
ECG_BNN [16]	62.49	91.90	67.75	81.99	17.93	78.72	51.54	65.58	<u>92.58</u>	<u>98.68</u>	<u>92.63</u>	93.24
ATI-CNN [17]	76.63	94.46	82.54	87.04	23.22	79.34	54.53	68.43	94.76	72.87	93.80	95.21
ACNet [18]	58.15	91.41	74.35	83.62	15.46	70.64	68.26	64.96	91.26	98.52	91.34	92.02
MVMSNet [19]	75.00	94.99	81.67	86.85	31.63	84.88	<u>57.69</u>	<u>72.93</u>	93.30	98.26	93.39	93.80
Ours	84.62	96.44	87.32	91.03	33.45	87.23	<u>58.37</u>	74.59	96.52	99.50	96.44	96.90

4.3. Implementation details

We provide the architectural configurations of the proposed network and experiment configurations in Table 3. Dim indicates the length of the patch dimension in patch embedding. Heads indicates the number of heads in the Transformer encoder. Ratio is the expanding ratio of the feed-forward network in the Transformer encoder. Layers represents the number of layers of CNN Blocks and Transformer encoder in the network. Our experiments are performed using the PyTorch framework. The optimization of the network is achieved by utilizing AdamW, while the learning rate is decayed by a cosine annealing decay strategy. Our experiments are performed on a single RTX 2080-Ti GPU.

4.4. Evaluation metrics

To ensure effective evaluation of network performance, it is crucial to employ reasonable evaluation metrics. We select the area under the curve (AUC), F1 score, recall, and accuracy as evaluation metrics for reflecting network performance in the Chapman dataset. Since accuracy is not a suitable evaluation indicator in the multi-label dataset, we use mean average precision (mAP) instead of accuracy. The mAP takes into account the precision and recall values for each label and computes the average across all labels. This metric provides a more comprehensive evaluation of the network's performance in a multi-label classification task. Therefore, for multi-label datasets like the CPSC dataset and the PTB-XL dataset, we employ AUC, F1 score, recall, and mAP as evaluation metrics.

5. Results and discussion

5.1. Comparison with previous methods

This experiment aims to verify the superior performance of the proposed network. To comprehensively compare its superiority, we select several existing traditional deep learning methods for time series classification and ECG signal classification methods as baseline networks. To ensure consistency and comparability,

we apply the same data preprocessing method and research configuration to all methods across all three datasets.

A comparison of AUC, F1 score, recall, mAP, and accuracy across the networks is shown in Table 4, respectively. As shown in Table 4, our proposed network achieves the best performance across all evaluation metrics in almost all tasks on all three datasets. Our proposed network achieves 84.62% on the F1 score and 87.32% on the recall for the CPSC dataset, which outperforms the other networks for more than 2%. For the Chapman dataset, our proposed network achieves 96.90% on the accuracy, which outperforms the other methods by around 1.5%. Furthermore, for the PTB-XL dataset, almost all evaluation metrics are improved by a large degree compared to the other methods and only the recall of ACNet is higher than our proposed network. Our network achieves these superior performances at all three datasets, clearly demonstrating the benefits of its dual-branch structure for learning multi-scale features and effectiveness in detecting real diseases for ECG signal.

The proposed network also demonstrates generalizability across different datasets. For instance, resnet1d101 performs the best besides our proposed network on the CPSC dataset, but ECGNet performs better than resnet1d101 on the Chapman dataset. In addition, apart from our proposed network, InceptionTime shows the best performance on the PTB-XL dataset. Due to the generalization ability of the network, the performance of the network may vary with different datasets. A network that performs well on one dataset may not necessarily perform equally well on another. These experiments demonstrate the effectiveness of our network for different ECG datasets. From Fig. 6, we can observe that as the number of iterations increases, all four metrics (AUC, F1 score, recall, and accuracy) of the training data gradually improve on the CPSC dataset. This indicates that the network is learning from the training data and making progress in terms of classification performance. During the first 75 training epochs, the training loss decreases rapidly, and then stabilizes as training progresses. This stabilization also suggests the convergence of the network, as the training loss no longer shows substantial fluctuations.

Table 5

The effect of CNN block on network performance. Bold numbers indicate the best performance.

Ablation	CPSC				PTB-XL				Chapman			
	F ₁	AUC	Recall	mAP	F ₁	AUC	Recall	mAP	F ₁	AUC	Recall	Accuracy
w/o CNN blocks	74.82	93.79	80.07	87.06	24.50	80.74	48.34	69.65	92.25	98.29	92.30	92.96
w/ CNN blocks(ours)	84.62	96.44	87.32	91.03	33.45	87.23	58.37	74.59	96.52	99.50	96.44	96.90

Table 6

Performance of different kernel sizes in dual-branch MSEL on all three datasets. Bold numbers indicate the best performance.

MSEL's kernel size		CPSC				PTB-XL				Chapman			
Upper branch	Lower branch	F ₁	AUC	Recall	mAP	F ₁	AUC	Recall	mAP	F ₁	AUC	Recall	Accuracy
20	5	83.00	96.28	85.83	90.35	30.70	84.08	58.83	72.09	96.07	99.38	96.09	95.59
40	5	83.38	96.15	86.01	90.70	29.37	88.01	56.77	72.87	95.29	99.52	95.30	96.53
20	100	81.04	95.26	84.57	89.70	30.97	84.66	54.58	70.44	94.96	99.36	94.99	95.40
40	100	83.26	95.28	86.01	90.49	31.65	85.14	56.23	72.38	95.40	99.20	95.53	95.59
(20, 40)	(5, 100)	84.62	96.44	87.32	91.03	33.45	87.23	58.37	74.59	96.52	99.50	96.44	96.90

Table 7

Performance of each branch network on all three datasets. Bold numbers indicate the best performance.

Branch	CPSC				PTB-XL				Chapman			
	F ₁	AUC	Recall	mAP	F ₁	AUC	Recall	mAP	F ₁	AUC	Recall	Accuracy
Upper	83.06	95.99	86.81	90.50	31.17	83.67	55.89	71.37	95.72	99.42	95.41	96.06
Lower	84.37	95.62	87.25	90.93	32.64	87.08	56.91	71.42	96.01	99.25	96.13	96.40
Ours	84.62	96.44	87.32	91.03	33.45	87.23	58.37	74.59	96.52	99.50	96.44	96.90

5.2. Ablation experiments

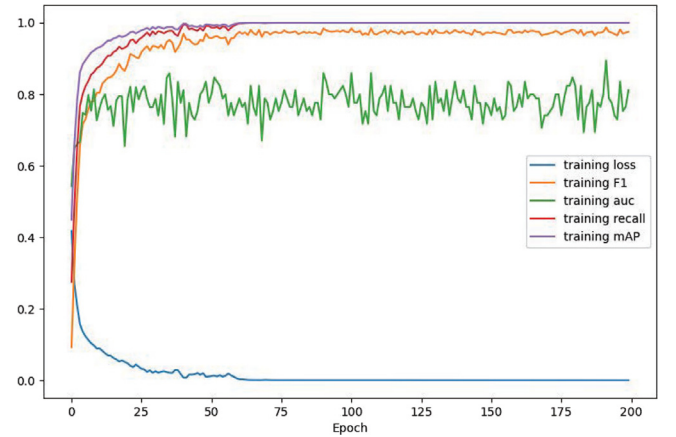
In this section, we conduct ablation experiments on the above three datasets to evaluate the impact of different components in our proposed network. Each component is eliminated individually by retraining the model while keeping other structures in the network unchanged. The goal is to gain a comprehensive understanding of each component's contribution to the overall performance of the network.

5.2.1. Ablation of CNN blocks

To reveal the influence of the CNN blocks in our proposed network, we conduct an ablation study to compare the performance difference of our network with and without CNN blocks. As shown in Table 5, the CNN blocks provide a large improvement. When the CNN blocks are removed from the network, we can see that all metrics in the three datasets drop to varying degrees. These results show that applying convolutions with different kernel sizes in parallel helps the network to effectively enhance feature representation.

5.2.2. Multi-Scale Embedding Layer (MSEL) analysis in dual-branch

We compare the effect of multi-scale embedding layers with conventional single-scale embedding layers on our proposed network performance on the ECG signal classification tasks in all three datasets. By using different kernel sizes in the single-scale embedding layers of dual-branch, we investigate the impact of capturing different scales of ECG features on improving the network's performance. As shown in Table 6, almost all evaluation metrics of our proposed network are improved when replacing the single-scale embedding layers with the MSELs, which indicates that the multi-scale feature learning used in the MSEL performs well in ECG signal classification. In Table 6, the results demonstrate that the kernel size used in the embedding layer has a significant impact on the performance of our proposed network. Specifically, when the kernel size of the lower branch MSEL is 1×100 , the 1×40 kernel in the upper branch MSEL brings more than 1.4% absolute improvement for the F1 score and recall in the CPSC dataset compared with the 1×20 kernel. This analysis helps us understand the importance of selecting an appropriate kernel size to effectively capture scale features in ECG signal and improve the overall performance of the proposed network.

**Fig. 6.** Loss and metrics of our proposed network during training.

5.2.3. Ablation of multibranch

To evaluate the contribution of each branch, we evaluate the ECG classification performance of each branch compared to our proposed dual-branch structure network. Table 7 summarizes the results. As observed from the results in Table 7, both our upper branch network and lower branch network exhibit decreased ECG classification performances in comparison with our proposed network. This highlights the impact of the dual-branch structure in our proposed network. The two-branch configuration enables the network to learn complementary information across distinct time scales. Moreover, the classification performance of the lower branch network outperforms that of the upper branch network. This observation might be attributed to the fact that cross-scale feature information acquired at more disparate scales is more conducive to enhancing network performance.

5.2.4. Ablation studies about the TS block

(1) *Ablation Studies about η* : The η represents the overall proportion of patch tokens that are retained after the selection process in the TS block. This parameter determines the amount of information that is preserved and re-fed into the Transformer

Table 8Performance of different values for η on all three datasets. Bold numbers indicate the best performance.

η	CPSC				PTB-XL				Chapman			
	F ₁	AUC	Recall	mAP	F ₁	AUC	Recall	mAP	F ₁	AUC	Recall	Accuracy
0	83.38	96.30	86.16	90.66	32.01	87.19	57.41	73.57	94.70	98.96	94.73	95.12
0.75	84.06	96.41	86.96	90.78	30.04	86.85	56.88	71.82	95.43	99.61	95.19	95.77
1	84.27	96.35	86.88	90.41	34.93	86.74	58.30	73.60	95.78	99.37	95.78	96.15
0.5(default)	84.62	96.44	87.32	91.03	33.45	87.23	58.37	74.59	96.52	99.50	96.44	96.90

Table 9

Performance of different ratios in the negative token selection on all three datasets. Negative represents to arrange tokens from high to low according to the global class attention score, and selects tokens in negative order. Bold numbers indicate the best performance.

Ratio (%)	CPSC				PTB-XL				Chapman			
	F ₁	AUC	Recall	mAP	F ₁	AUC	Recall	mAP	F ₁	AUC	Recall	Accuracy
0%(negative)	83.38	96.30	86.16	90.66	32.01	87.19	57.41	73.57	98.96	94.73	95.12	94.70
30%(negative)	83.86	96.53	87.32	90.64	30.69	87.76	59.89	72.95	95.57	99.40	95.60	96.53
50%(negative)	82.75	95.65	83.99	89.50	33.32	87.76	61.55	74.05	96.46	99.41	96.42	96.81
Ours(positive)	84.62	96.44	87.32	91.03	33.45	87.23	58.37	74.59	96.52	99.50	96.44	96.90

encoders. By conducting experiments with different values of η and comparing the results, the study aims to evaluate the impact of retaining different proportions of patch tokens on the performance of the proposed network. Table 8 presents the results of the proposed network with different values of η on all three datasets. The $\eta = 0$ and $\eta = 1$ indicate that no patch tokens are re-fed to Transformer encoders and all patch tokens are re-fed to Transformer encoders, respectively. From the table, it can be observed that when setting $\eta = 0.5$, the network achieves improved performance compared to other values of η . Overall, the results indicate that discarding an appropriate proportion of unnecessary or redundant patch tokens in the TS block can effectively reduce information redundancy in ECG signals. By removing redundant information, the network can focus more on the informative and discriminative features, leading to improved performance in all ECG signal classification tasks in all three datasets.

(2) *Negative Token Selection Analysis*: Negative global class attention refers to a mechanism to select patch tokens with lower global class attention scores. By conducting an experiment where patch tokens with lower global class attention scores are selected for re-input into the Transformer encoders, this study aims to evaluate whether retaining the highest score patch tokens outperforms the lowest score patch tokens. Table 9 shows the results in all evaluation metrics of our proposed network with negative global class attention on all three datasets. For instance, the ratio setting of 50% (negative) means to reserve the patch tokens with global class scores in the bottom 50%. We can see that, our proposed network which retains the patch tokens with global class scores in the top 50% outperforms the networks with 50% (negative) and 30% (negative) ratio settings. This suggests that prioritizing the retention of patch tokens with higher global class attention scores leads to improved performance in the proposed network. By focusing on the most informative and relevant patch tokens, the network can make better use of the attention mechanism and achieve better results in all ECG signal classification tasks in all three datasets.

(3) *Ablation Studies about λ* : The λ used in the global class attention is a smoothing factor used to estimate the local average of attention values in the encoders, so the update of attention values is related to their historical value across different encoders. Table 10 presents the results of our proposed network with different values of λ on all three datasets, evaluating the network's performance using various metrics. The results demonstrate that the inclusion of global class attention significantly improves the performance compared to the network without global class attention. The global class attention provides valuable information that helps the network better understand and utilize the attention

values. Since $\lambda = 0.99$ performs better overall than $\lambda = 0.9$, we set $\lambda = 0.99$ as the default value. In summary, the global class attention allows the network to better estimate the importance of patches and update attention values based on their historical values, leading to more accurate and reliable ECG signal classification results.

(4) *Ablation Studies about Weight Sharing*: The necessity of weight sharing for two sets of Transformer encoders in the TS block is verified through ablation experiments. The results in Table 11 demonstrate that without weight sharing, the performance metrics show a decline on all three datasets. This confirms the effectiveness of weight sharing in our proposed network. One possible explanation is that Transformer encoders with shared weights can more effectively extract useful features from selected high-importance patch tokens. Moreover, weight sharing helps reduce the total number of parameters in the model, improving parameter efficiency. Excessive parameter growth can lead to overfitting and increased computational demands, making weight sharing a beneficial strategy in our approach.

5.2.5. Ablation studies about the loss function

As shown in Eq. (6), we use $\mathcal{L}(Y, \text{CLASS})$ to make the output of the proposed network fit the truth label, and use $\mathcal{L}_{\text{FC}}$ to improve the quality of the extracted features. The results of ablation experiments in the loss function are shown in Table 12. In Exp. I, we do not use the feature correlation loss term $\mathcal{L}_{\text{FC}}$, and the result of the experiment indicate that the term $\mathcal{L}_{\text{FC}}$ plays a crucial role in improving the performance of the proposed network on the three datasets. In Exp. II, we also try to make the term $\mathcal{L}_{\text{FC}}$ in Eq. (6) to $\mathcal{L}_{\text{RR}}^{\text{A}}$, which can also improve the performance of most metrics on the three datasets. The result presented in Exp. II indicate that while the new loss function performs better than that in Exp. I, it performs poorly in comparison to the results in Exp. IV. In Exp. III, we replace $\mathcal{L}_{\text{FC}}$ in Eq. (6) with $1/\mathcal{L}_{\text{RR}}^{\text{B}}$ for research comparison, and find that half of the metrics decrease after replacement compared to the result in Exp. I. One possible reason is that $1/\mathcal{L}_{\text{RR}}^{\text{B}}$ not only helps to reduce redundant information in the CLS tokens of both branches, but also reduces the feature information that is beneficial to ECG signal classification. But the addition of $\mathcal{L}_{\text{RR}}^{\text{A}}$ helps to enhance the correlation of the two CLS tokens directly used for classification, thus preventing the reduction of feature information. The complementarity of $\mathcal{L}_{\text{RR}}^{\text{A}}$ and $1/\mathcal{L}_{\text{RR}}^{\text{B}}$ is also the reason why the results in Exp. IV performs better than that in Exp. II.

Table 10Performance of different values for λ on all three datasets. Bold numbers indicate the best performance.

λ	CPSC				PTB-XL				Chapman			
	F ₁	AUC	Recall	mAP	F ₁	AUC	Recall	mAP	F ₁	AUC	Recall	Accuracy
0	82.81	95.81	87.75	91.55	31.60	88.18	58.29	73.69	95.37	99.39	95.42	95.02
0.9	83.38	96.32	86.88	91.27	33.34	86.80	59.69	74.35	95.57	99.55	95.66	96.06
0.99(default)	84.62	96.44	87.32	91.03	33.45	87.23	58.37	74.59	96.52	99.50	96.44	96.90

Table 11

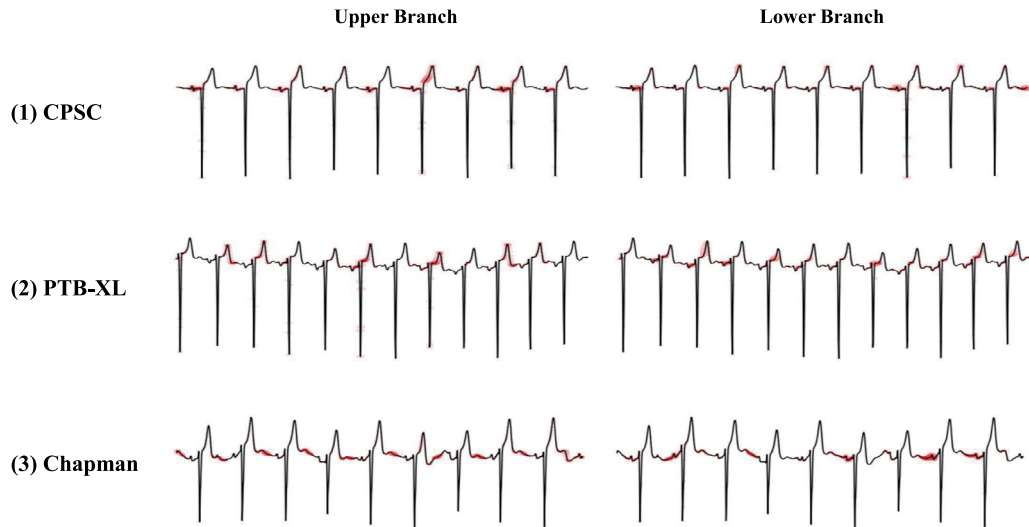
The effect of weight sharing on network performance. Bold numbers indicate the best performance.

Ablation	CPSC				PTB-XL				Chapman			
	F ₁	AUC	Recall	mAP	F ₁	AUC	Recall	mAP	F ₁	AUC	Recall	Accuracy
w/o weight sharing	82.73	96.14	86.45	90.71	32.41	86.56	58.94	74.02	95.72	99.58	95.70	96.15
w/ weight sharing(ours)	84.62	96.44	87.32	91.03	33.45	87.23	58.37	74.59	96.52	99.50	96.44	96.90

Table 12

Performance comparison between different loss function on all three datasets. Bold numbers indicate the best performance.

Ablation	CPSC				PTB-XL				Chapman			
	F ₁	AUC	Recall	mAP	F ₁	AUC	Recall	mAP	F ₁	AUC	Recall	Accuracy
I w/o $\mathcal{L}_{FC}$	83.95	95.75	85.51	90.26	34.93	86.74	58.30	73.60	95.90	99.20	95.91	96.24
II $\mathcal{L}_{FC} \rightarrow \mathcal{L}_{RR}^A$	84.55	96.40	87.11	90.77	33.15	86.79	58.97	74.00	96.05	99.48	95.97	96.53
III $\mathcal{L}_{FC} \rightarrow 1/\mathcal{L}_{RR}^B$	82.69	96.06	86.23	90.47	32.17	83.96	58.91	71.14	94.53	99.38	94.48	95.12
IV w/ $\mathcal{L}_{FC}$ (ours)	84.62	96.44	87.32	91.03	33.45	87.23	58.37	74.59	96.52	99.50	96.44	96.90

**Fig. 7.** Visualization of ECG features based on the two branches in our proposed network. Since the ECG is one-dimensional signal, we represent the attention intensity for each part of the ECG using the size of red dots instead of a heatmap.

5.3. ECG feature visualization

Gradient-weighted Class Activation Mapping (Grad-CAM) has been employed as a visual explanation tool to interpret and understand the performance of Deep Neural Networks [36]. The Grad-CAM can generate heatmaps that highlight the important regions of the input data that influence the network's predictions. In this paper, we employ this method to visualize ECG features from the two branches, separately. Fig. 7 illustrates feature visualizations of one lead of the ECGs in all three datasets. The brighter color represents more attention to the corresponding ECG signal. In the proposed network, attentions in both branches are directed towards the same location in the ECG signal. This means that both branches of the network focus on the same features in the input ECG data that are critical for classification performance. The proposed network ensures that both branches effectively contribute to ECG signal classification, exploiting their

respective strengths and capturing the most relevant information for accurate prediction.

6. Limitations

While the proposed network demonstrates substantial improvements in the task of ECG signal classification, it is important to acknowledge several limitations in this study. First, our network accepts an unsegmented 12-Lead ECG signal containing multiple heartbeats. Therefore, our proposed method mainly focuses on capturing the heartbeat features in ECG signal. However, there are other important features in ECG signal classification, such as ST segment, QT interval, etc., which may not be fully considered in our method. Another limitation is that each branch in a multi-branch input requires a different backbone. This results in a significant consumption of hardware resources.

7. Conclusion

In this paper, we propose a novel network for ECG signal classification that leverages a dual-branch structure and effectively learns multi-scale features. In our network, the dual-branch structure and multi-scale embedding layers are integrated to obtain different time scale features. Meanwhile, the combination of CNN and Transformer enables our network to capture both local and global information, leading to a more comprehensive understanding of the underlying patterns in the signal. Our network also explores the influence of patch token selection and introduces the concept of global class attention. By selectively choosing patch tokens based on their global class attention scores, we have enhanced the network's ability to capture relevant feature information and discard redundant information. A feature correlation loss function is also presented to mitigate the information redundancy problem. These approaches are shown to be effective in reducing information redundancy in ECG signal and improving network performance. Research results show our proposed network can achieve competitive results than the existing works.

In future work, we will pay more attention to solving the information redundancy problem for the ECG signal classification and develop proper multi-scale feature extraction technology for 12-lead ECG signal. In addition, we use the basic Transformer encoder in our proposed network. We hope to explore the structure of the Transformer encoder to further improve our network performance.

CRedit authorship contribution statement

Siyuan Zhang: Writing – original draft, Software, Methodology, Conceptualization. **Cheng Lian:** Supervision, Project administration, Funding acquisition, Formal analysis. **Bingrong Xu:** Writing – review & editing. **Junbin Zang:** Writing – review & editing, Investigation. **Zhigang Zeng:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported in part by the Natural Science Foundation of China under Grants 62176193 and 62206204; in part by the Key R&D Program of Shanxi Province, China (No. 202102020 101010).

References

- [1] G.A. Mensah, G.A. Roth, V. Fuster, The global burden of cardiovascular diseases and risk factors, *J. Am. Coll. Cardiol.* 74 (20) (2019) 2529–2532.
- [2] J.A. Nasiri, M. Naghibzadeh, H. Yazdi, et al., ECG arrhythmia classification with support vector machines and genetic algorithm, in: 2009 Third UKSim European Symposium on Computer Networking and Simulation, IEEE, 2009, pp. 187–192.
- [3] G. Goovaerts, S. Padhy, B. Vandenberg, et al., A machine-learning approach for detection and quantification of QRS fragmentation, *IEEE J. Biomed. Health Inform.* 23 (5) (2019) 1980–1989.
- [4] Awni Y. Hannun, P. Rajpurkaret, et al., Cardiologist-level arrhythmia detection and classification in ambulatory electrocardiograms using a deep neural network, *Nature Med.*, (1) 65–69.
- [5] P. Schwab, G.C. Scebbia, J. Zhang, et al., Beat by beat: Classifying cardiac arrhythmias with recurrent neural networks, in: 2017 Computing in Cardiology (CinC), IEEE, 2017, pp. 1–4.
- [6] P. Xie, G. Wang, C. Zhang, et al., Bidirectional recurrent neural network and convolutional neural network (BiRCNN) for ECG beat classification, in: 2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), IEEE, 2018, pp. 2555–2558.
- [7] G. Yan, S. Liang, Y. Zhang, et al., Fusing transformer network with temporal features for ECG heartbeat classification, in: 2019 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), IEEE, 2019, pp. 898–905.
- [8] W. Wang, W. Chen, Q. Qiu, et al., CrossFormer++: A versatile vision transformer hinging on cross-scale attention, 2023, arXiv preprint arXiv:2303.06908.
- [9] C. Lai, S. Zhou, N.A. Trayanova, Optimal ECG-lead selection increases generalizability of deep learning on ECG abnormality classification, *Philos. Trans. R. Soc. A* (2212) (2021) 20200258.
- [10] C.K. Jha, M.H. Kolekar, Tunable q-wavelet based ecg data compression with validation using cardiac arrhythmia patterns, *Biomed. Signal Process. Control* 66 (2021) 102464.
- [11] J.A. Gutiérrez-Gnechchi, R. Morfin-Magana, D. Lorias-Espinoz, et al., DSP-based arrhythmia classification using wavelet transform and probabilistic neural network, *Biomed. Signal Process. Control* 32 (2017) 44–56.
- [12] M. Llamado, J.P. Martínez, An automatic patient-adapted ECG heart-beat classifier allowing expert assistance, in: 2012 IEEE Transactions on Biomedical Engineering, Vol. 59, pp. 2312–2320.
- [13] Ö. Yildirim, A novel wavelet sequence based on deep bidirectional LSTM network for ECG signal classification, *Comput. Biol. Med.* 96 (2018) 189–202.
- [14] H.M. Lynn, S.B. Pan, P. Kim, A deep bidirectional GRU network for biometric electrocardiogram classification based on recurrent neural networks, *IEEE Access* 7 (2019) 145395–145405.
- [15] B. Murugesan, V. Ravichandran, K. Ram, et al., ECGNet: Deep network for arrhythmia classification, in: 2018 IEEE International Symposium on Medical Measurements and Applications (MeMeA), IEEE, 2018, pp. 1–6.
- [16] N. Pu, Z. Wu, A. Wang, et al., Arrhythmia classifier based on ultralightweight binary neural network, 2023, arXiv preprint arXiv:2304.01568.
- [17] Q. Yao, R. Wang, X. Fan, et al., Multi-class arrhythmia detection from 12-lead varied-length ECG using attention-based time-incremental convolutional neural network, *Inf. Fusion*, 53, 174–182.
- [18] T. Chen, C. Huang, E. Shih, et al., Detection and classification of cardiac arrhythmias by a challenge-best deep learning neural network, *iScience* 23 (3) (2020).
- [19] S. Yang, C. Lian, Z. Zeng, et al., A multi-view multi-scale neural network for multi-label ECG classification, *IEEE Trans. Emerg. Top. Comput. Intell.* (2023).
- [20] H. Han, C. Lian, Z. Zeng, et al., Multimodal multi-instance learning for long-term ECG classification, *Knowl.-Based Syst.* 270 (2023) 110555.
- [21] H.I. Fawaz, B. Lucas, G. Forestier, et al., Inceptiontime: Finding alexnet for time series classification, *Data Min. Knowl. Discov.* 34 (6) (2020) 1936–1962.
- [22] A. Trockman, J.Z. Kolter, Patches are all you need? 2022, arXiv preprint arXiv:2201.09792.
- [23] A. Dosovitskiy, L. Beyer, A. Kolesnikov, et al., An image is worth 16 × 16 words: Transformers for image recognition at scale, 2022, arXiv preprint arXiv:2010.11929.
- [24] M. Chen, M. Lin, K. Li, et al., A general coarse-to-fine method for vision transformer, 2022, p. 1, arXiv preprint arXiv:2203.03821.
- [25] F. Liu, C. Liu, L. Zhao, et al., An open access database for evaluating the algorithms of electrocardiogram rhythm and morphology abnormality detection, *J. Med. Imaging Health Inform.* 8 (7) (2018) 1368–1373.
- [26] P. Wagner, N. Strodthoff, R.D. Boussejot, et al., PTB-XL, a large publicly available electrocardiography dataset, *Sci. Data*, 7, (1) 154.
- [27] J. Zhang, J. Zhang, S. Danioko, et al., A 12-lead electrocardiogram database for arrhythmia research covering more than 10,000 patients, *Sci. Data* 7 (1) (2020) 1–8.
- [28] Z. Wang, W. Yan, T. Oates, Time series classification from scratch with deep neural networks: A strong baseline, in: Proc. Int. Joint Conf. Neural Networks, 2017, pp. 1578–1585.
- [29] K. He, X. Zhang, S. Ren, et al., Deep residual learning for image recognition, in: Proc. IEEE Conf. Comput. Vis. Pattern Recognit., 2016, pp. 770–778.
- [30] T. He, Z. Zhang, H. Zhang, et al., Bag of tricks for image classification with convolutional neural networks, in: Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR), 2019, pp. 558–567.

- [31] H.I. Fawaz, B. Lucas, G. Forestier, et al., InceptionTime: Finding AlexNet for time series classification, *Data Min. Knowl. Discov.* 34 (2020) 1936–1962.
- [32] C. Szegedy, S. Ioffe, V. Vanhoucke, A.A. Alemi, Inceptionv4, inception-resnet and the impact of residual connections on learning, in: *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2017, pp. 4278–4284.
- [33] A. Howard, M. Sandler, G. Chu, et al., Searching for mobilenetv3, in: *Proc. IEEE Int. Conf. Comput. Vis.*, 2019, pp. 1314–1324.
- [34] S. Hochreiter, J. Schmidhuber, Long short-term memory, *Neural Comput.* 9 (8) (1997) 1735–1780.
- [35] S. Zhang, D. Zheng, X. Hu, et al., Bidirectional long short-term memory networks for relation classification, in: *Conf. PACLIC*, 2015, pp. 73–78.
- [36] G.B. Moody, R.G. Mark, The MIT-BIH arrhythmia database on CD-ROM and software for use with it, in: *Proc. Comput. Cardiol.*, 1990, pp. 185–188.